Ysbyty Gwynedd
Anaesthetics Department
Adran Anaestheteg

CRITICAL CARE

DIABETIC EMERGENCIES

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Author(s): Eirlys Walker
Reviewer: Dr Will Sutcliffe

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Key Points

- Include Hypoglycaemic events, DKA, Hyperosmolar Hyperglycaemic State (HSS) and Euglycaemic DKA (EDKA)
- Should be classed as medical emergencies
- Hypo boxes are available on the wards for patients
- To be read in conjunction with YG CG 1007 'Diabetic Ketoacidosis'

HYPOGLYCAEMIA

Defined as a blood sugar <4.0mmol/L, Hypoglycaemia is the commonest side effect in the treatment of diabetes. It can be split by severity in to:

MILD - where the BM is low, but the patient is able to self-treat.

SEVERE – where the intervention of a third party is required to resolve the situation.



Locate the nearest HYPO BOX which contains: Glucotabs, Gucojuice and Glucagon

TREATMENT

Treatment is dependent on the patient's ability to swallow and conscious level.

CONSCIOUS, ORIENTATED AND ABLE TO SWALLOW	CONSCIOUS BUT CONFUSED, UNABLE TO COOPERATE BUT ABLE TO SWALLOW.	SEMI-CONSCIOUS OR UNCONSCIOUS PATIENT (UNSAFE SWALLOW)
1. Give 15-20g of fast acting carbohydrate *	1. Give 1 ½ - 2 tubes of 40% glucose gel (squeezed between the teeth and gums) OR 1mg of glucagon IM (once only)	1. If the patient has IV access: give: 75-100 mls of 20% glucose over 15 mins OR 150-200 mls of 10% glucose over 15 mins
2. Repeat capillary blood glucose measurement 10-15 mins later. If still <4mmol/L, repeat step 1 above up to 3 times	2. Repeat BM in 10-15 mins. If BM is still <4.0mmol/L repeat glucose gel up to 3 times.	2. If the patient doesn't have IV access give 1mg of glucagon IM (once only) ***
3. If BM remains <4.0mmol/L after 30-45 mins or 3 cycles, escalate and consider: • 1mg Glucagon IM (once only) • 150-200mls 10% Glucose IV	3. If BM remains low give 150-200mls of 10% glucose IV	3. If BM remains low repeat the IV treatment above up to 3 times. (Consider 10% Glucose infusion if BM remains low)
4. Once the BM .4.0mmol/L and the patient has recovered, give 20g long acting carbohydrate (40g if the patient has had glucagon) **	4. Once BM is >4.0mmol/L give 20g of long acting carbohydrate (40g if patient has had glucagon) **	4. Once BM is >4.0mmol/L give 20g of long acting carbohydrate (40g if patient has had glucagon) **

* e.g. 4-5 Glucotabs; 60ml Glucojuice; 150-200ml pure fruit juice; 3-4 heaped tsp of sugar dissolved in water.

**20g= e.g. 2 biscuits; 1 slice bread/toast; 200-300ml glass of milk; next meal if due.

*** In patients who are malnourished, have a history of alcohol abuse or severe liver disease IV glucose is preferred.

For all of the above – if the hypo was due to sulfonylurea or long acting insulin risk of hypo may persist for 24-36 hrs. Consider IV 10% glucose infusion.

HYPEROSMOLAR HYPERGLYCAEMIC STATE (HSS)

Formally known as HONK (hyperglycaemic hyperosmolar non-ketotic), HSS is a serious condition. It is distinct and separate from DKA but is still and emergency. It normally affects older people, but may be seen in teenagers and adolescents. It is often the first presentation of T2 diabetes.

CHARACTERISTIC FEATURES:

- Hypovolaemia (often severe)
- Marked hyperglycaemia >30mmol/L
- Raised osmolarity ≥320mosmol/Kg
- No significant ketonaemia (<3mmol/L)
- No significant acidicosis (pH >7.3, Bicarb >15)

BCUHB_Guidelines_Mgmt_HHS_Adults-[

JBDS_IP_HHS_Adults.pdf

*NB: Some patients can present with a mixed HSS/DKA – if so seek help

TREATMENT

0-60 MINS	60 MINS – 6 HRS	6 – 12 HRS	12 – 24 HRS	24 HRS – 3 DAYS
- Start 0.9% Sodium chloride infusion 1L over 1 hr (consider faster rate if SBP <90; caution in elderly) - ONLY start Insulin infusion IF significant ketonaemia (>1mmol/l) -AIMS: Blood Glucose ↓ 5mmol/L/hr Osmolality ↓ 3-8 mosmol/kg/hr -MONITORING: BMs hrly, Osmolality hrly (calc using 2Na+Gluc+Urea) -Comm LMWH (unless contraindicated)	- Continue 0.9% Sodium Chloride infusion 0.5-1L/hr Target: 2-3L +ve fluid balance by 6hrs AIMS: Blood Glucose ↓ 5mmol/L/hr Osmolality ↓ 3-8 mosmol/kg/hr - Replace potassium as needed - Keep BM 10-15mmol/l If BM <14 comm 5% - 10% Dextrose infusion alongside Sodium Chloride Measure BMs and Osmolality hrly	- Continue 0.9% Sodium Chloride infusion. Target: 3-6 L +ve fluid balance by 12hrs - BMs hrly, Osmolality 2hrly - Continue IV glucose if needed to keep BM 10-15mmol/l - If no improvement – senior review - Diabetic team review - Assess for complications eg Cerebral oedema	- Continue IV fluids. Aim to replace estimated fluid losses - If Na and Osmolality improving measure 4 hrly, if not continue 2hrly measurement - BMs hrly - If no improvement - senior review - Assess for complications eg Cerebral oedema	- Continue IV fluids until eating and drinking normally - VRII if not eating and drinking - Subcutaneous insulin regimen if eating and drinking normally - Assess for complications eg Cerebral oedema - Continue LMWH until ready for discharge. - Diabetic Team review.

← Commence FRII (0.05 Unit/kg/hr) if BM plateaus →

EUGLYCAEMIC DIABETIC KETOACIDOCIS (EDKA)

EDKA will present with a picture of DKA ie Ph <7.3, Bicarb <15 and Ketones >3mmol/l but will have a normal, or only minimally elevated BM. It can happen with Type 1 or Type 2 Diabetes. Conditions associated with EDKA include pregnancy, high alcohol intake, insulin pump use, chronic liver disease, pancreatitis, reduced calorie intake and glycogen storage disorders. It presents a diagnostic challenge to clinicians due to the range of aetiologies and the normal blood glucose and can result in a delay in diagnosis. There has been an increase in the incidence of EDKA recently with the introduction of Sodium-Glucose Transporter 2 (SGLT2) inhibitors to treat Diabetes particularly in the first few weeks of starting the drug.

There is no guidance as yet from JBDS, but the American Diabetes Association and literature suggest:

Step 1: Initial fluid replacement with crystalloid 1-1.5L in the first hour in order to combat dehydration

Step 2: Commence Fixed Rate Insulin Infusion at a rate of 0.05-0.1 units/kg/hr to resolve the ketosis with a concurrent infusion of Dextrose 5% or 10% to prevent Hypoglycaemia.